BYU STEM Camp

RC Car Assembly & Engineering Guide

A Step-by-Step Hands-on Tutorial

Inventory: 12 Essential Components

Logic & Control

1x ESP32-CAM (OV2640)
1x DRV8833 Controller
2x 8-Pin Female to Male Headers
1x 6-Pin Male to Male Headers

Drivetrain

4x TT DC Motors
4x Rubber Wheels
4x T-Brackets
8x Screws

Power & Chassis

1x Custom PCB
1x 4 AA Battery Holder
1x Power Switch
4x AA Batteries
1x Foam or cardboard rectangle
1x Zip tie

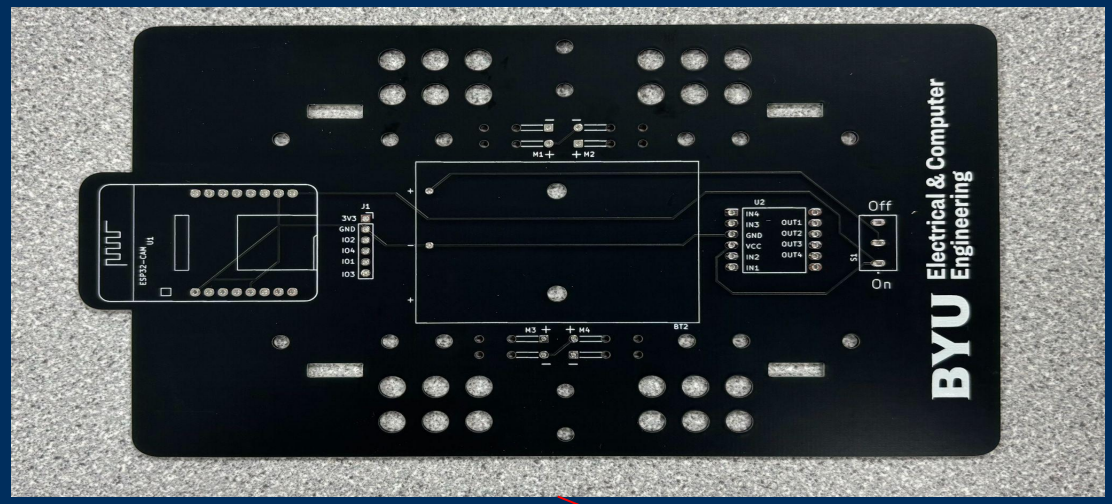
Phase 1: Precision Soldering

Establishing the Electrical Backbone

Preparing the PCB and Power Switch

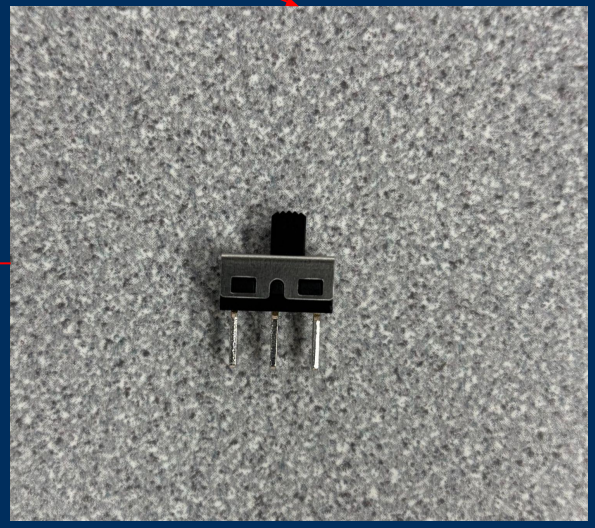
Steps 1-2: Base Setup

- **Initialize:** Secure your PCB in a safe, static-free workspace.
- **Switch Installation:** Insert the power switch into the designated footprint.
- **Quality Check:** Ensure the switch sits flush against the board before soldering the terminals tight.









Integrating the DRV8833 Controller

Step 3: Align Pins

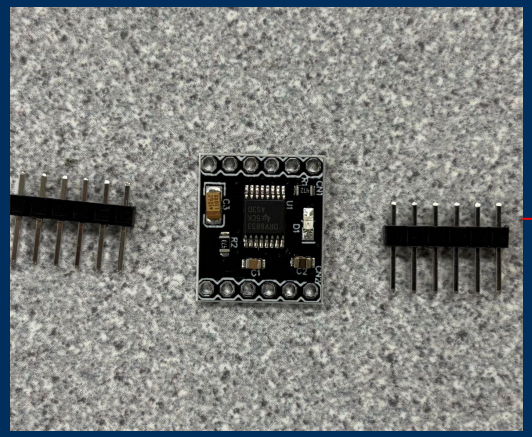
Insert two 6-pin headers. Ensure plastic spacers are on the top side. Align controller to the drawing.

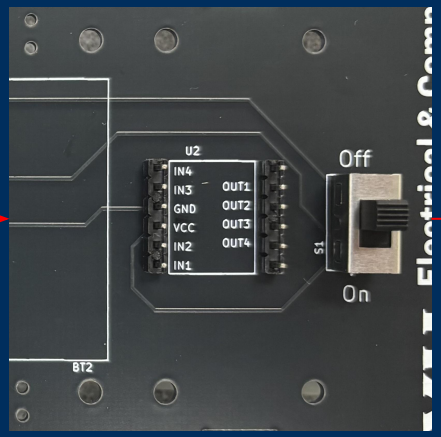
Step 4: Controller

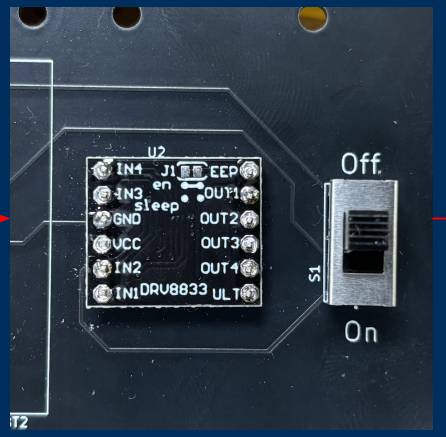
If needed, use a rubber band to hold the controller. Solder two anchor pins first, verify they are flush, then solder the remaining pins.

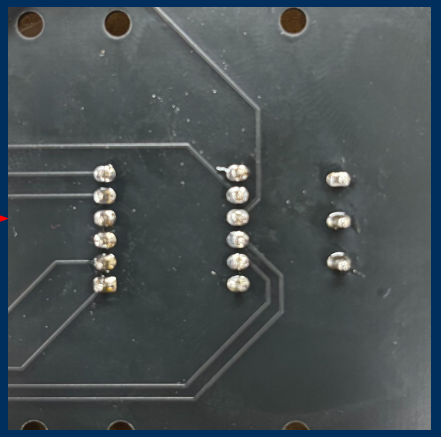
Step 5: Header Pins

Solder the two 6-pin headers that you just attached to the controller onto the PCB. Solder two anchor pins first, verify they are flush, then finish all the remaining pins.









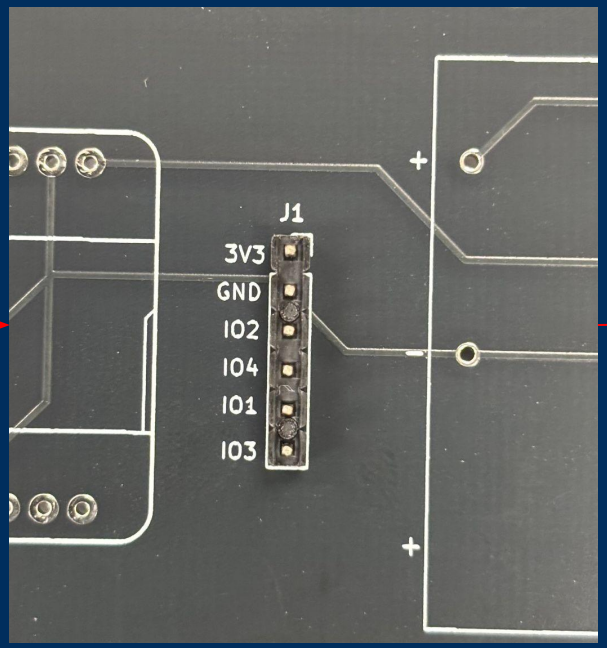
Lone Pins Installation

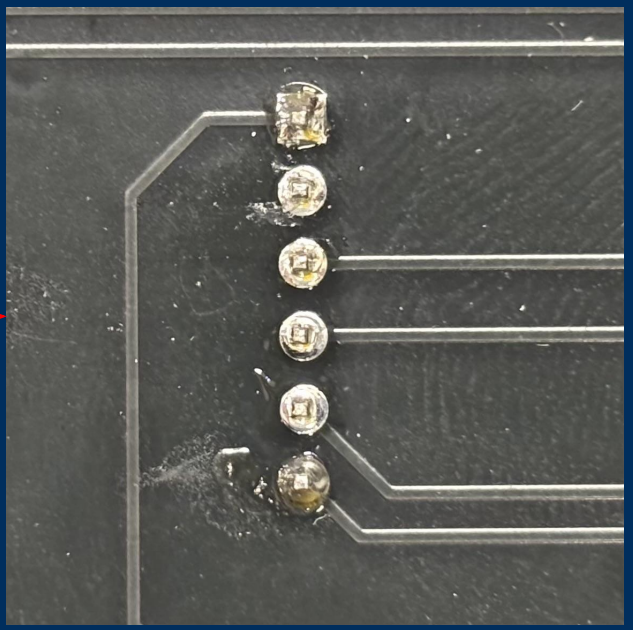
Step 6: Precise Alignment

Insert the 6-pin headers into the PCB. Ensure the plastic spacers are on the top side. Solder two anchor pins first, verify they are flush, then solder the remaining pins.

Be careful when touching the pins, as they will be hot!







Integrating the ESP32 Microcontroller

Step 7: Align Pins

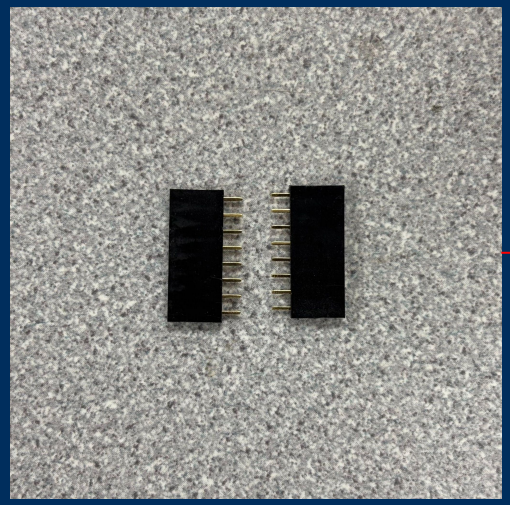
Insert the two 8-pin female header pins. Ensure plastic holes are on the top side. Align ESP32 to the drawing.

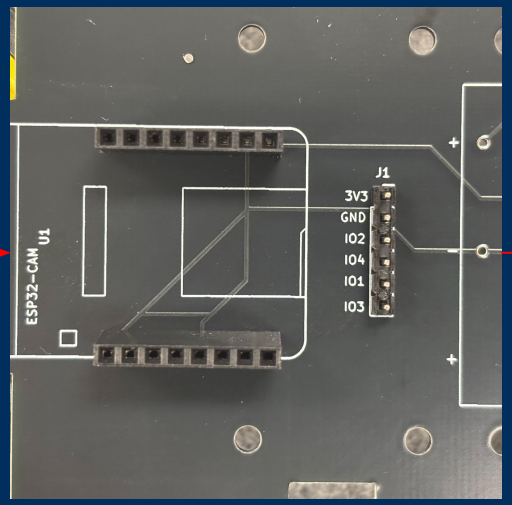
Step 8: Microcontroller

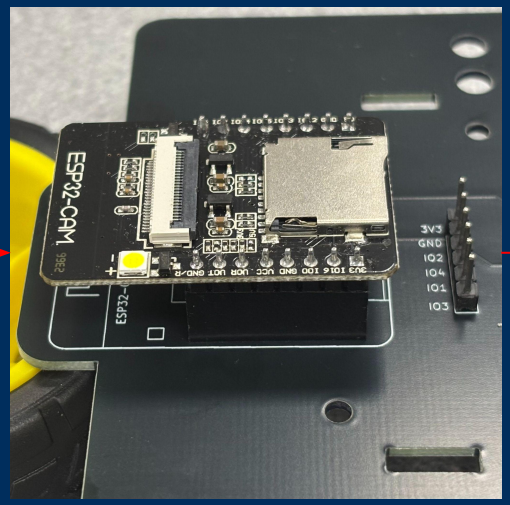
If needed, use a rubber band to hold the microcontroller. Solder two anchor pins first, verify they are flush, then solder the remaining pins.

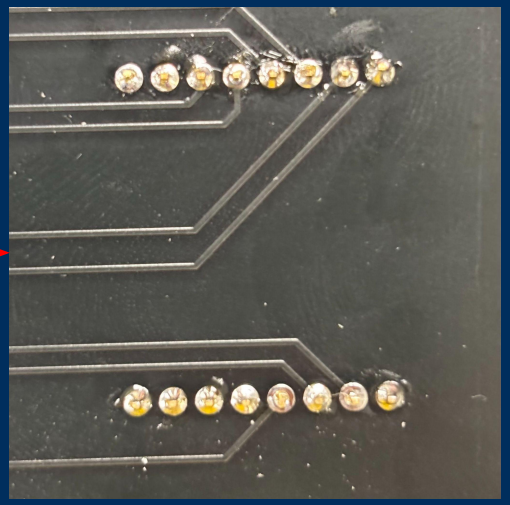
Step 9: Unplug the ESP32

Do not solder the ESP32 to the board. Remove it now, and place it in a safe location.





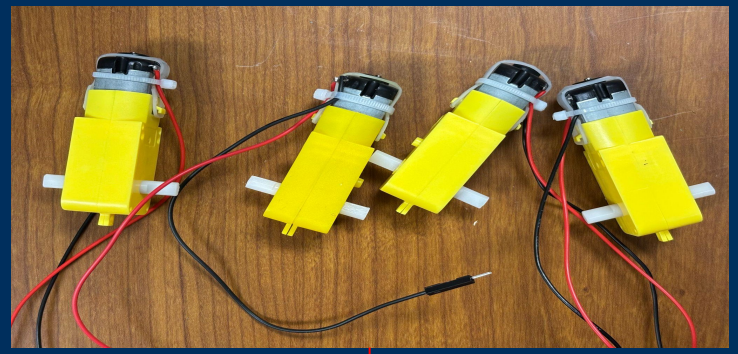


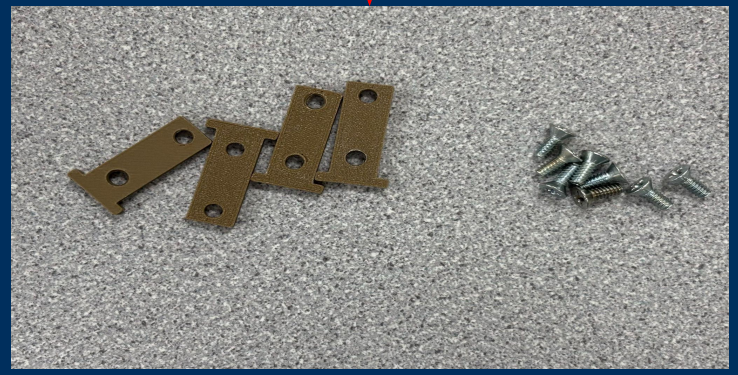


Phase 2: Mechanical Assembly

Constructing the Chassis and Drivetrain

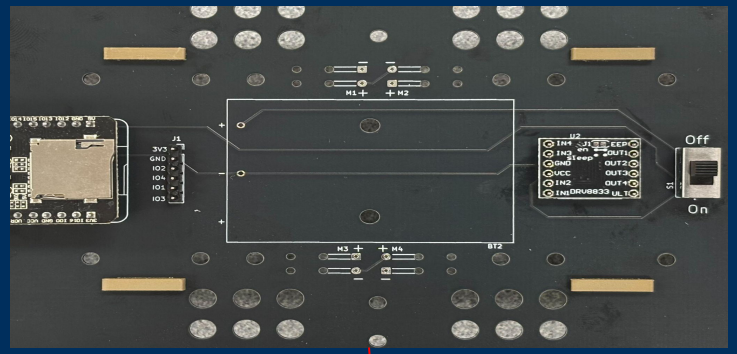
Mounting Motors and T-Brackets

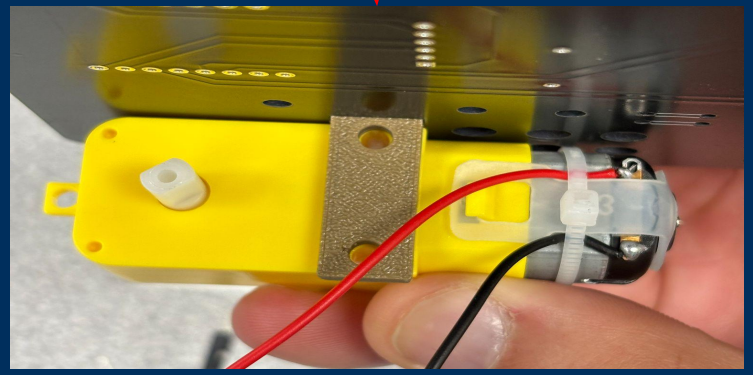




Step 10: Mount Prep

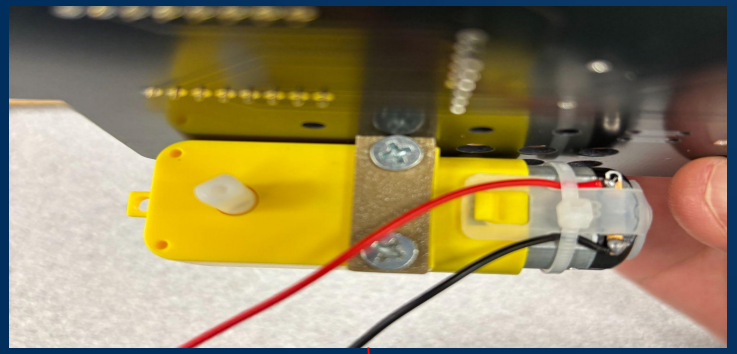
Gather all 4 motors, 4 T-brackets, and 8 screws.

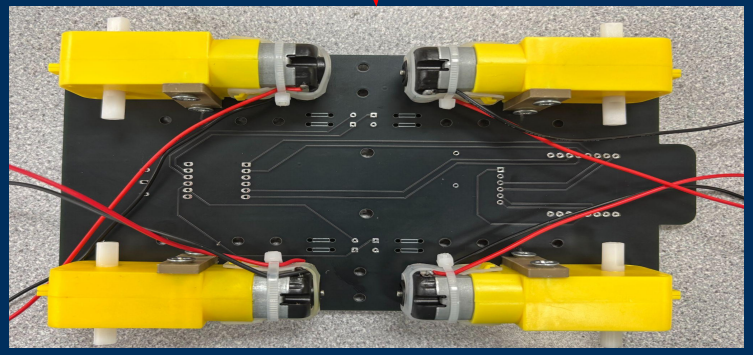




Step 11: Mount & Alignment

Place the T-brackets in their respective holes, then align the motor's holes with the holes in the T-bracket.





Step 12: Screw in & Wire Check

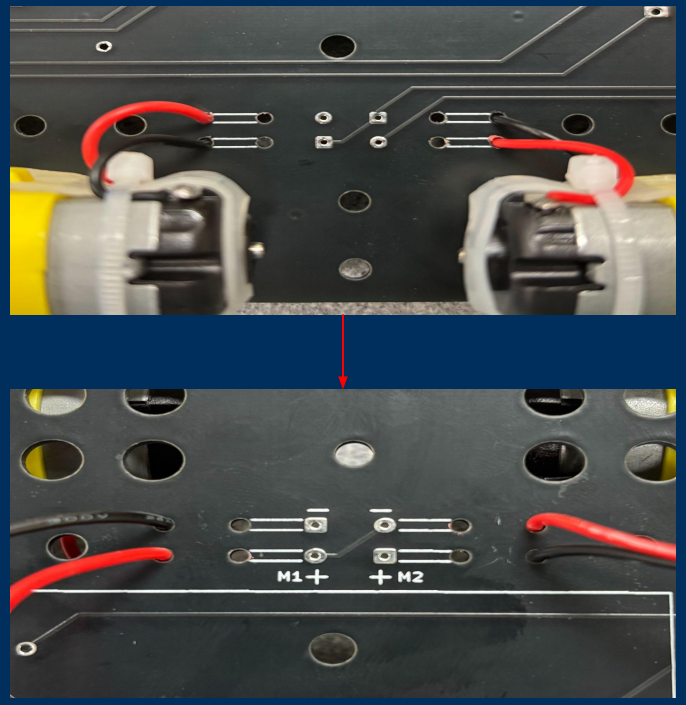
Insert two screws into the aligned holes and tighten them. Ensure the wires point inward to prevent snagging.

Critical Wiring Configuration

Correct polarity is essential for synchronized movement. Follow this mapping exactly:

Motor	PCB Label	Red Wire (+)	Black Wire (-)
M1 (Top Left)	M1	Positive (+)	Negative (-)
M2 (Top Right)	M2	Negative (-)	Positive (+)
M3 (Bottom Left)	M3	Positive (+)	Negative (-)
M4 (Bottom Right)	M4	Negative (-)	Positive (+)

Wires Routing & Polarity

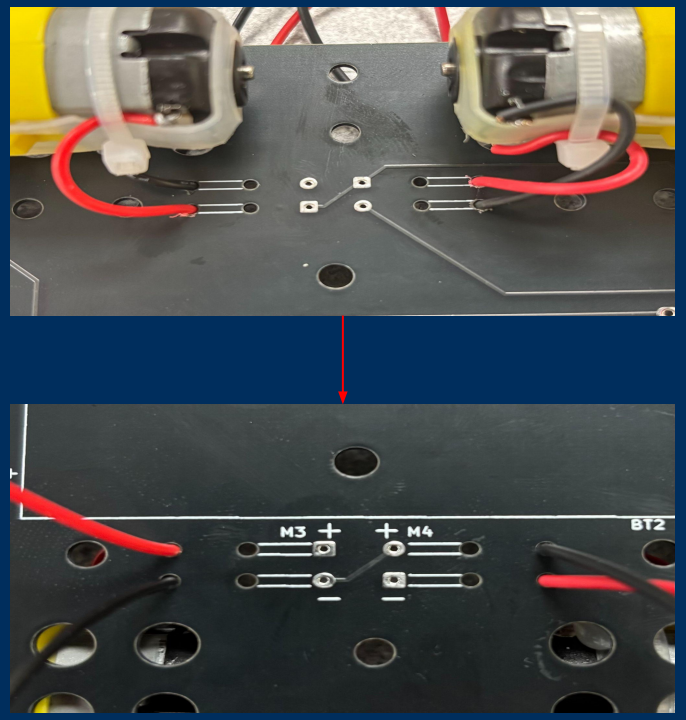


Step 13: Motors M1 & M2

Locate the "M1+" and "+M2" labels on the top side of the PCB (M1 is top-left; M2 is top-right). Thread the wires through the routing holes and connect them exactly as follows to ensure correct wheel rotation:

Motor M1: Red wire to (+) | Black wire to (-)

Motor M2: Black wire to (+) | Red wire to (-)



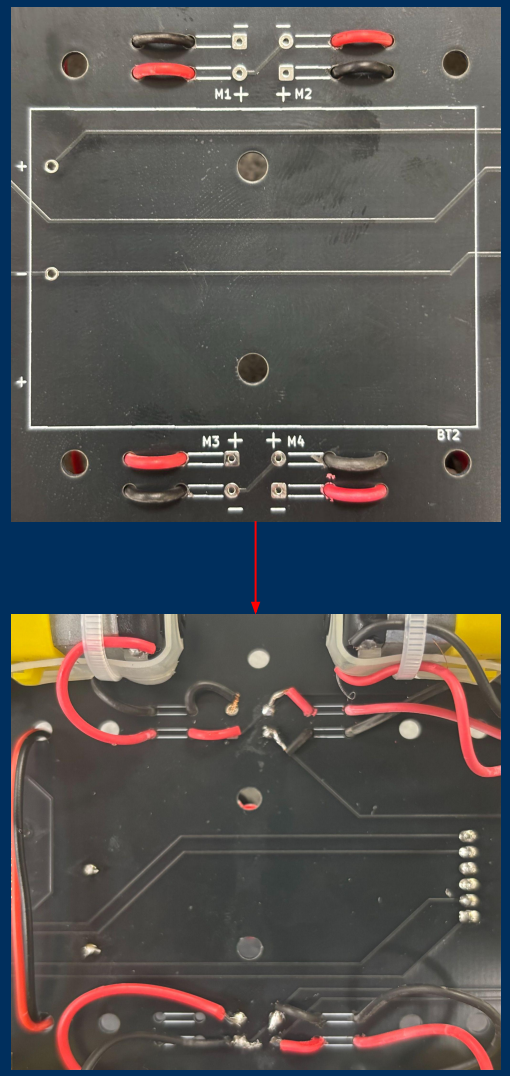
Step 14: Motors M3 & M4

Locate the "M3+" and "+M4" labels on the bottom side of the PCB (M3 is bottom-left; M4 is bottom-right). Thread the wires through the routing holes and connect them exactly as follows to ensure correct wheel rotation:

Motor M3: Red wire to (+) | Black wire to (-)

Motor M4: Black wire to (+) | Red wire to (-)

Final Motor Soldering & Routing



Step 15-18: Check & Solder

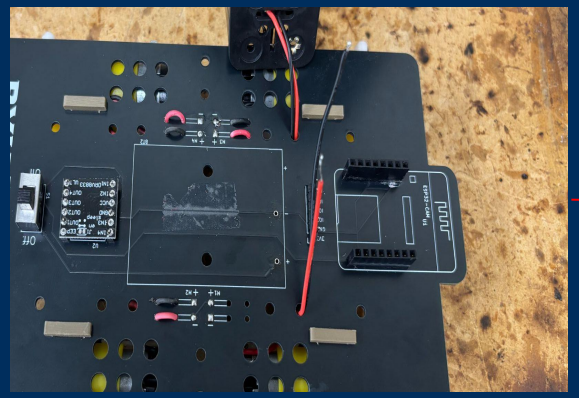
- **Routing:** Pass the wires through the small cable management holes.
- **Soldering:** Secure the wires to their corresponding square and circle pads on the PCB.
- **Verification:** Ensure the wires form straight lines and avoid leaving wire loops near the center of the board. Pull any excess wire toward the motors so that any remaining slack sits next to the motors rather than in the center. This strain relief ensures your solder joints remain durable over time.

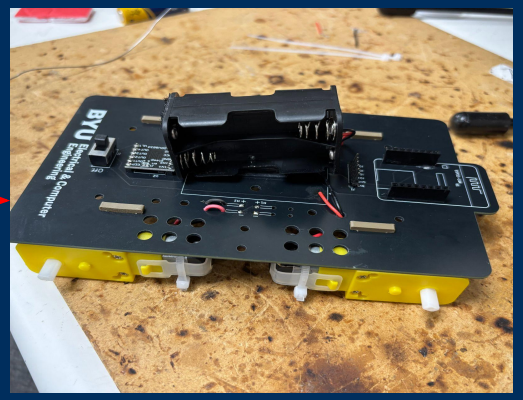
Power System Integration

Step 19–20: Wiring & Soldering

Route battery wires through hole, as shown in the image.

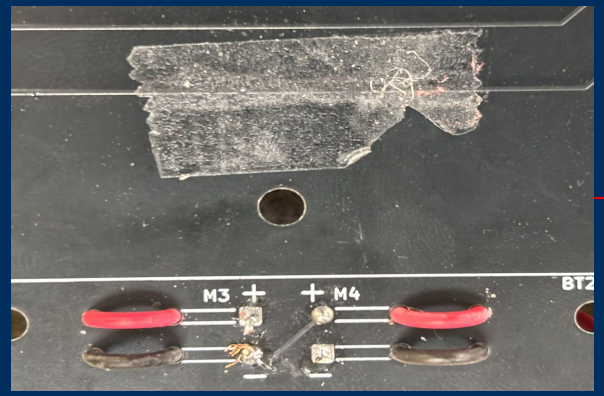
Solder Red to (+) and Black to (-) terminals.

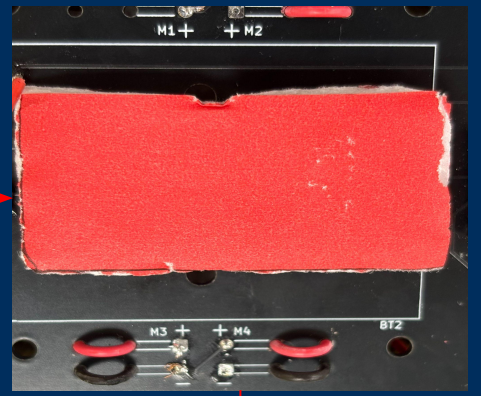


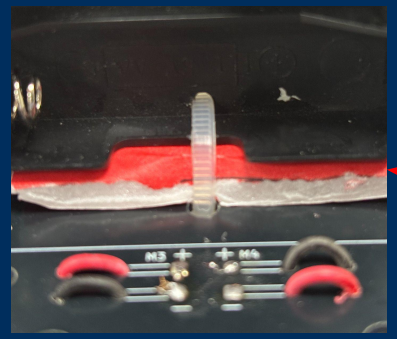


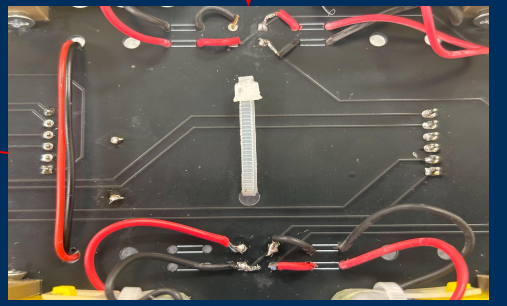
Step 21–22: Mounting

Apply the double-sided tape and red foam. Secure the 4xAA battery pack tightly using zip ties. Ensure the locking head of each zip tie sits underneath the PCB. You may use pliers to tighten the zip ties. Cut the excess.

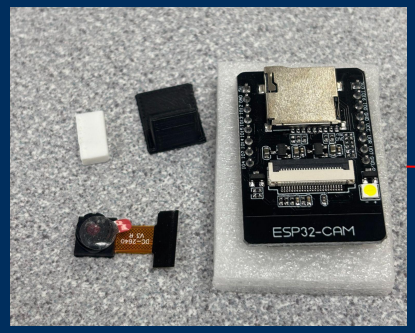


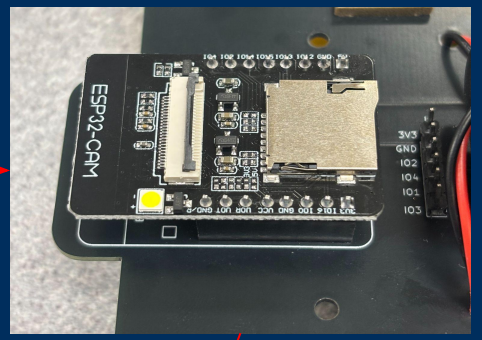


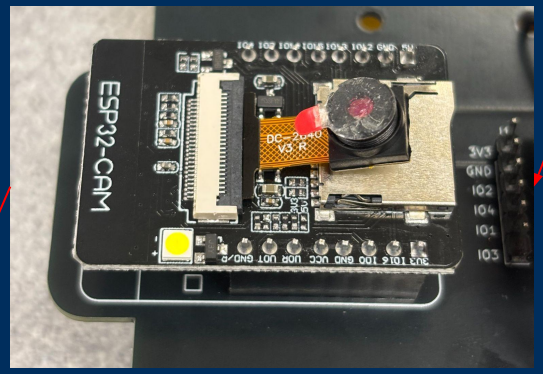


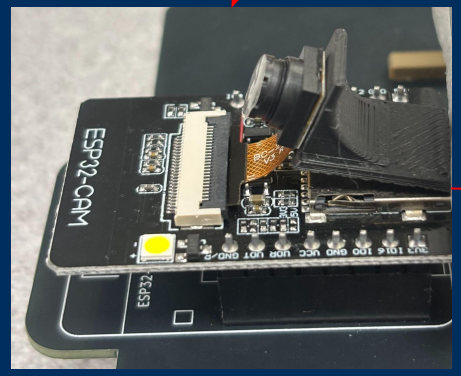


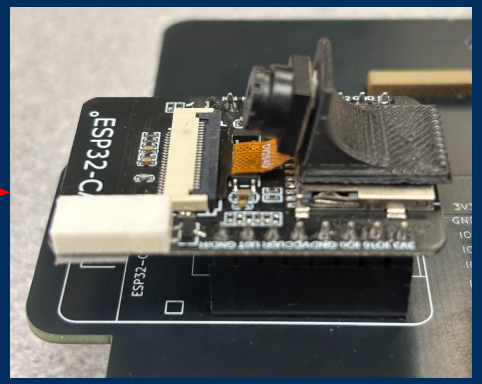
Final ESP32 Mounting











Step 23-25: Mount and Adjust

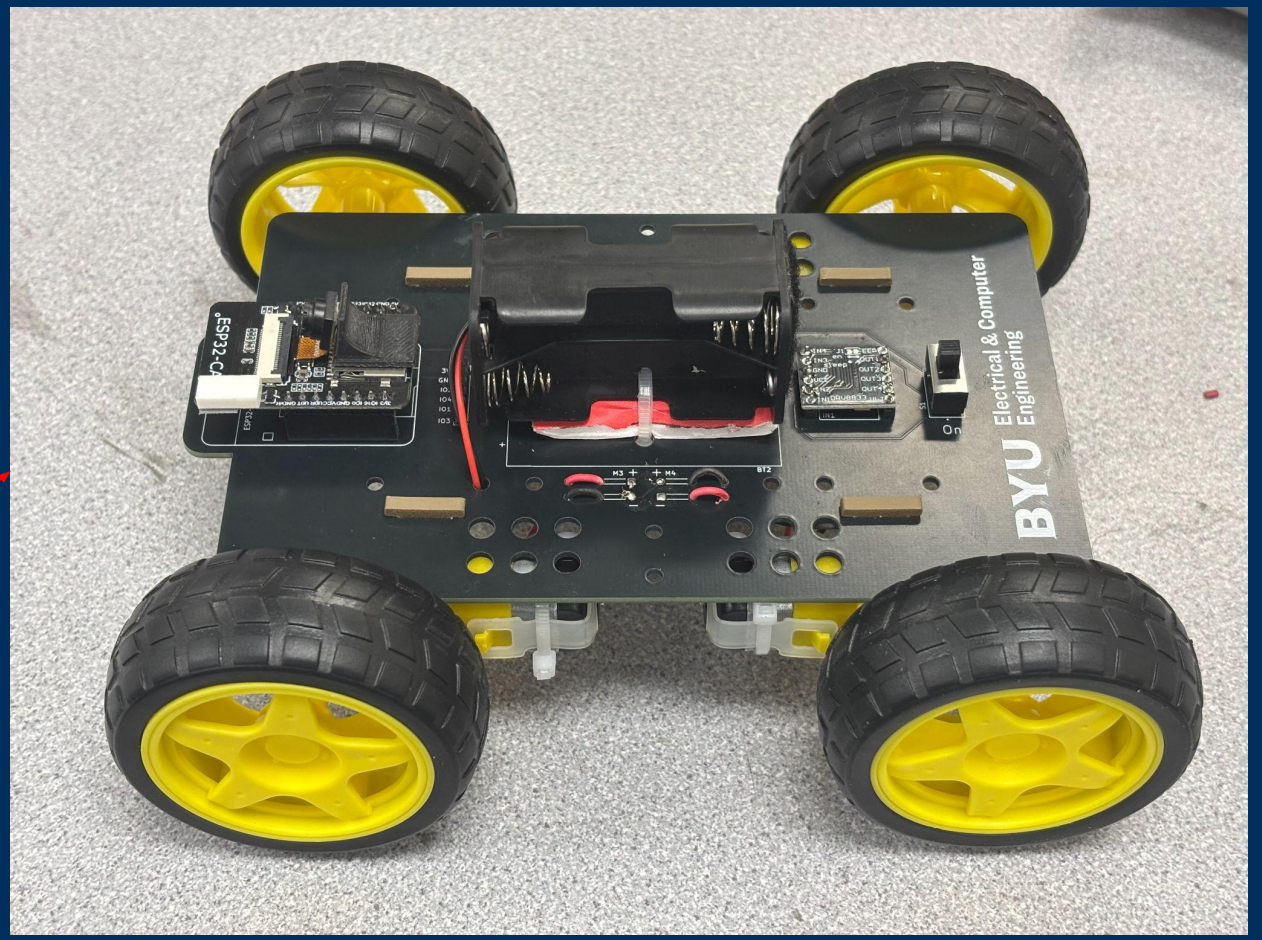
- **Mounting:** Ensure the power switch is turned off. Insert the ESP32 into the two 8-pin female headers.
- **Camera:** Insert the small camera module into the ESP32 as shown in the image below.
- **Camera Stand:** Apply a small piece of double-sided tape to the camera stand, then secure it onto the MicroSD card slot.
- **Headlight mount:** Locate the small, white rectangular headlight mount. Apply a small square of double-sided tape around the LED on the ESP32. Mount the headlight bracket by aligning its square hole with the LED, pressing it down to secure it with the tape.

Wheels Integration

Step 26: Wheel Mounting

Mount all four wheels onto the four motors. Once completed, your car should look like the image below.





Phase 3: Microcontroller Programming

Constructing and Testing the Brain of the Car

Ready for Inspection?



Step 27: STOP

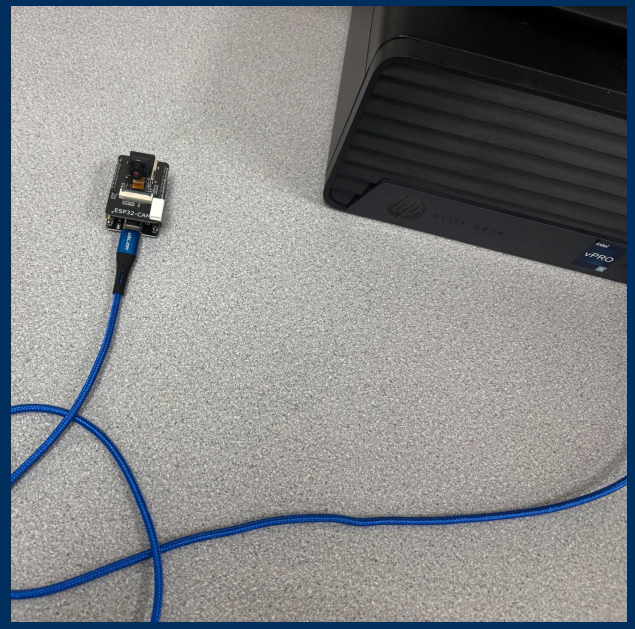
CAUTION: Do not install the batteries or turn on the power switch yet!

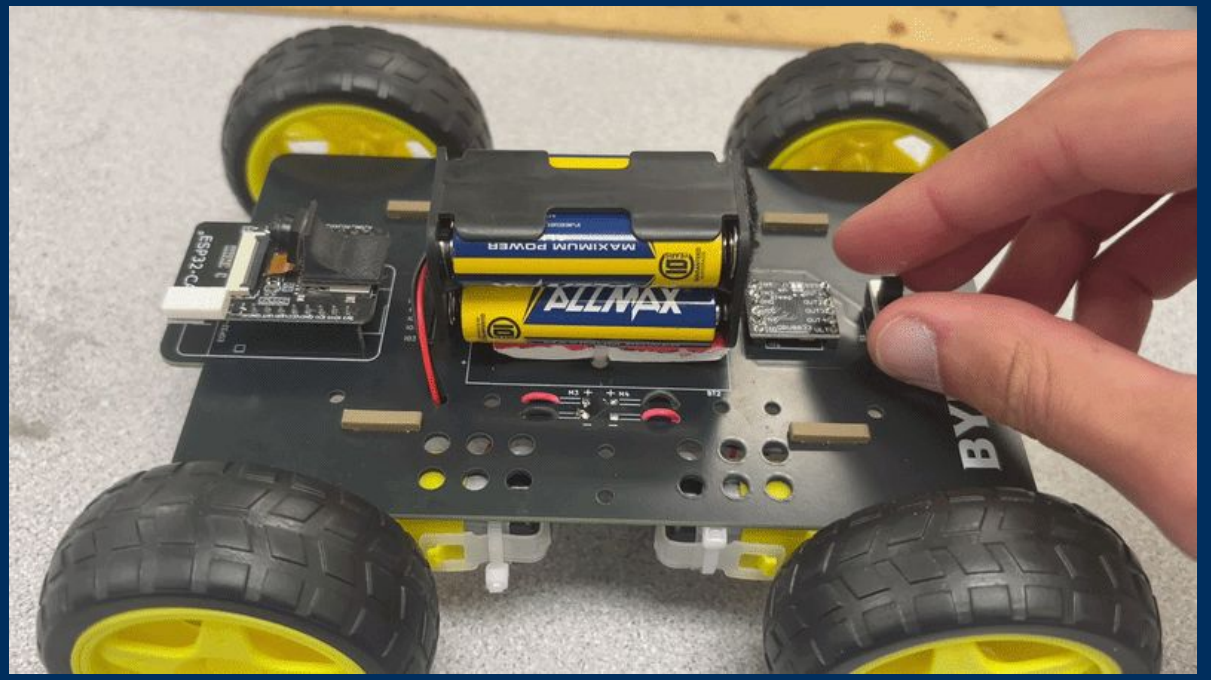
Please ask a **Lab Assistant** for a final check of your soldering and wiring before powering it on.

ESP32 Programming

Step 28: ESP32 Programming

If your car passes inspection, ask a lab assistant for help programming your ESP32 so it is ready to drive. During the programming process, ensure you assign a custom name to your ESP32's Wi-Fi network. The lab assistant will then flash your ESP32. If the upload is successful, the onboard LED will flash three times. Once programmed correctly, your ESP32 is ready to run, and you can now install the batteries.

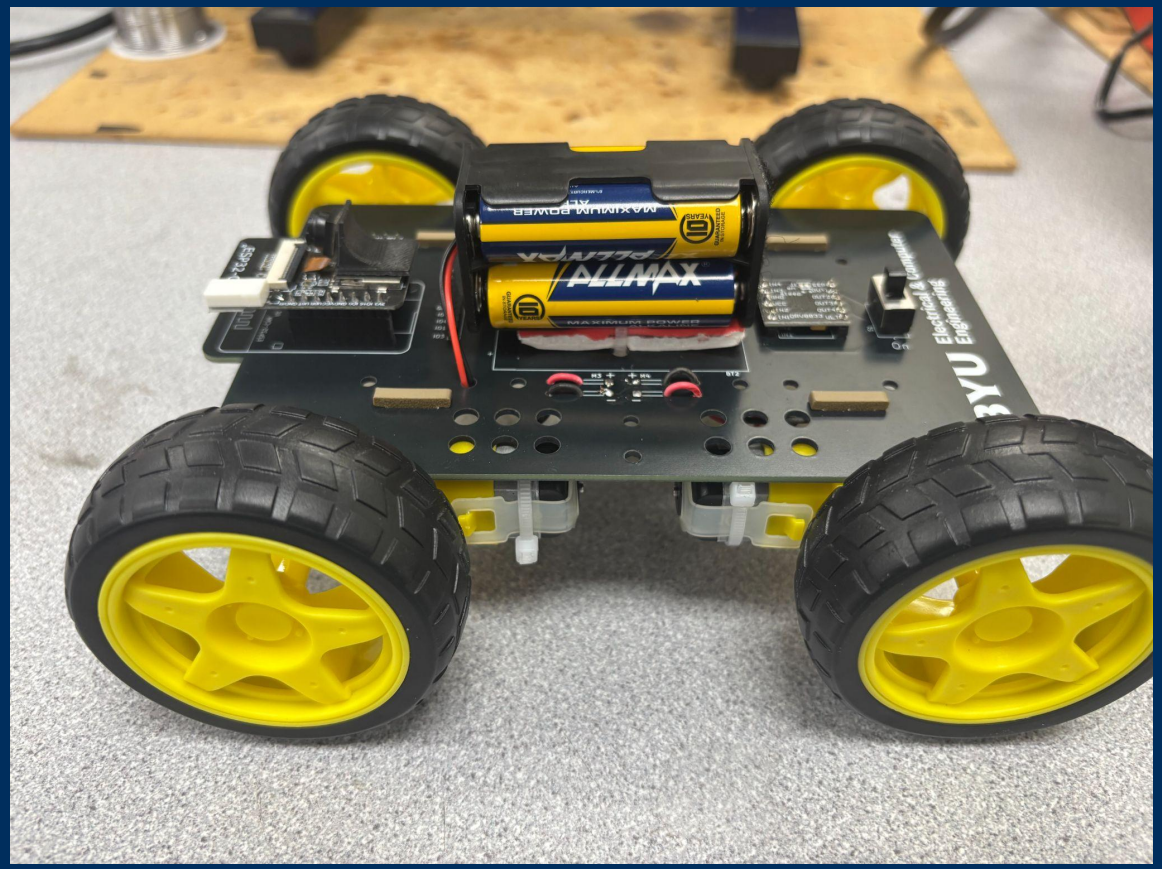




Battery Installation

Step 29: Battery Installation

Once you install the batteries, turn on the power switch. Verify that the motor controller LED turns on and that the ESP32 LED flashes three times. This indicates that your motor controller is receiving power and the ESP32 has booted up correctly.

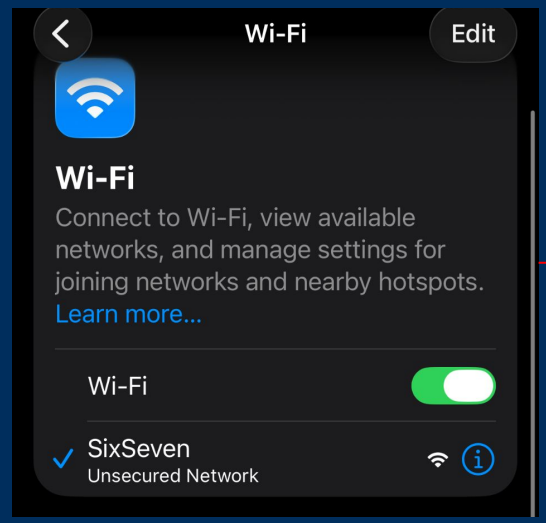


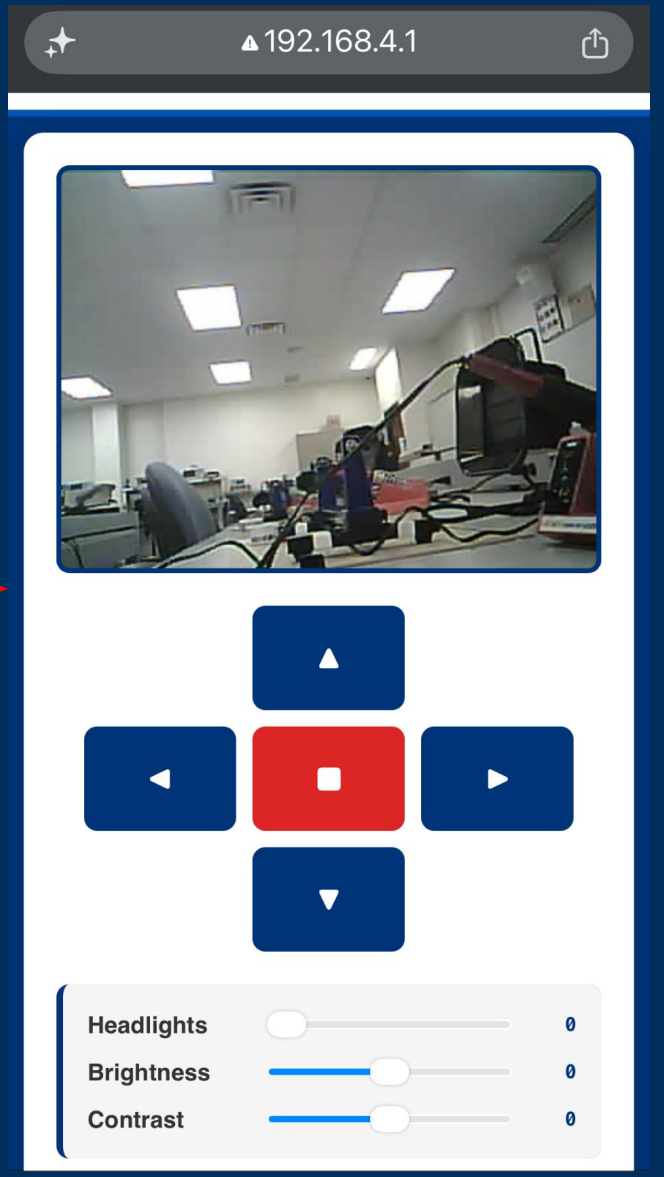
Wi-Fi Connection

Step 30: Connecting to your car

Now that your car has power and the ESP32 has booted up correctly, open the Wi-Fi settings on your smartphone or computer. Look for your custom network name and connect to it (for example, we called it “SixSeven”). This is an open network, so no password is required. Next, open a standard web browser (such as Chrome or Safari), type the following IP address into the URL bar, and press enter:

http://192.168.4.1/



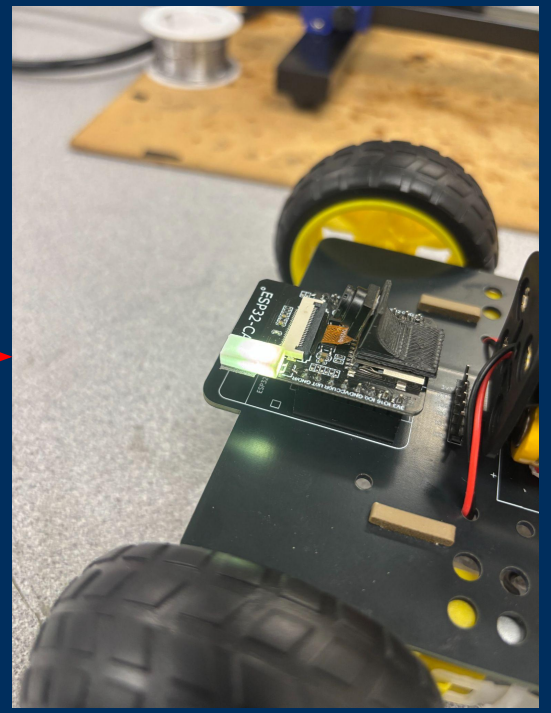


Using the controls

Step 31: Learning the controls interface

The dashboard interface provides full control over the ESP32-CAM car, featuring a live video feed, directional pads for movement, and a brake button. It also includes sliders for adjusting the physical LED headlights, camera sensor exposure, and video contrast for better navigation. For more details, visit the guide's documentation.



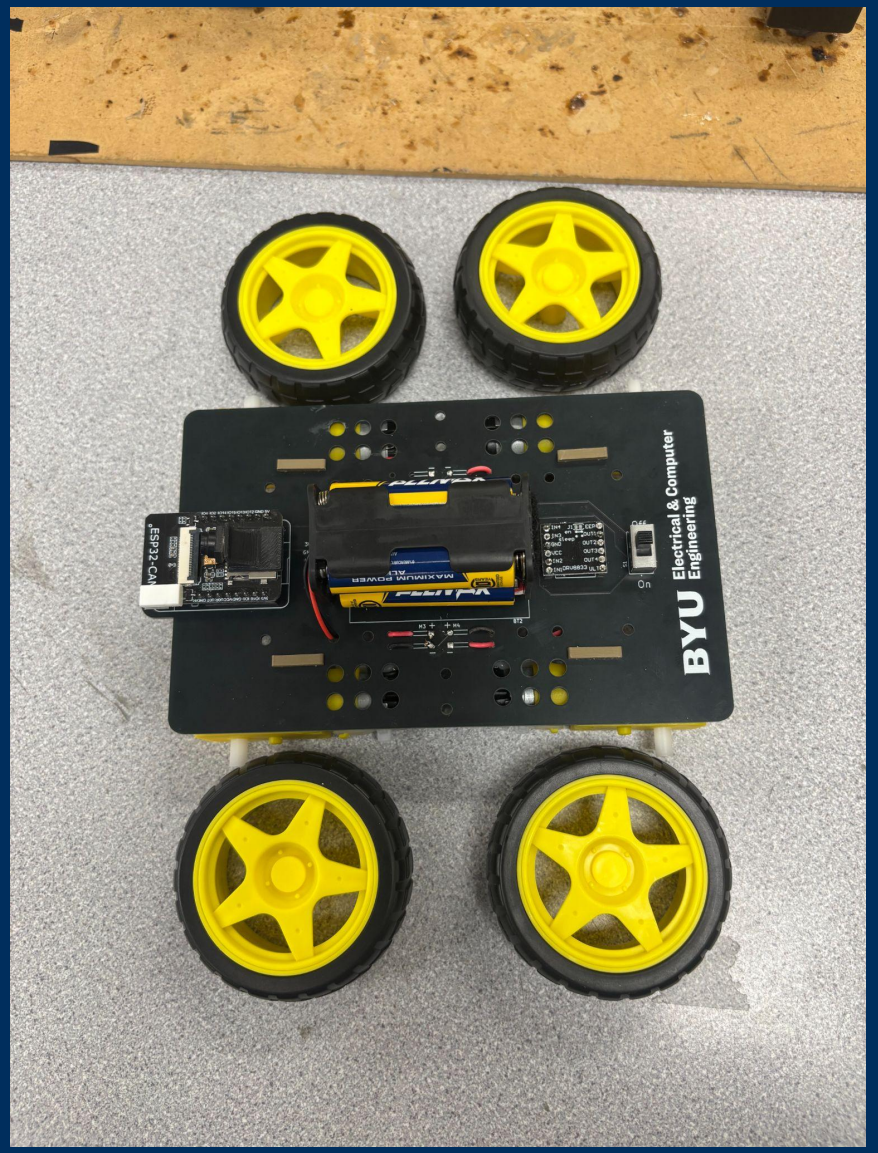


Packaging and Storage

Step 32: Take home

When you're done testing and need to pack up your car for storage or transport (especially in a suitcase), follow these steps:

- **Power off:** Make sure your car is turned off.
- **Remove the Wheels:** Detach the wheels from the motor shafts to prevent damage during transport.
- **Pack Carefully:** Once the wheels are removed, your car should fit easily in a suitcase or storage container.



Ready for the race?



Congratulations!

You have successfully completed the step-by-step assembly guide.

Now go enjoy your car, and have a friendly race with your friends. Let fastest win!

Troubleshooting & More

Some troubleshooting steps:

If the live video freezes or you lose control of the car, you have likely driven out of Wi-Fi range. Try these steps to fix it:

- **Get Closer:** Walk back toward the car so your phone is within a clear line of sight, and refresh the web page.
- **Restart the Wi-Fi:** Turn the Wi-Fi off and then back on in your phone or computer settings, then reconnect to your car's network.
- **Restart the Car:** Turn the car's power off, wait a few seconds, and turn it back on. Reconnect your phone to the network and reload the dashboard.

Keep developing at home:

The best part of engineering is breaking things and making them your own. If you want to see exactly how this code works or start modifying the dashboard and motor controls yourself, you can find the complete source code online.

Go to https://github.com/vrsp05/STEM-Camp-RC-Car.git to view the project. Be sure to Fork the repository to your own account so you can start making your own changes and improvements!